

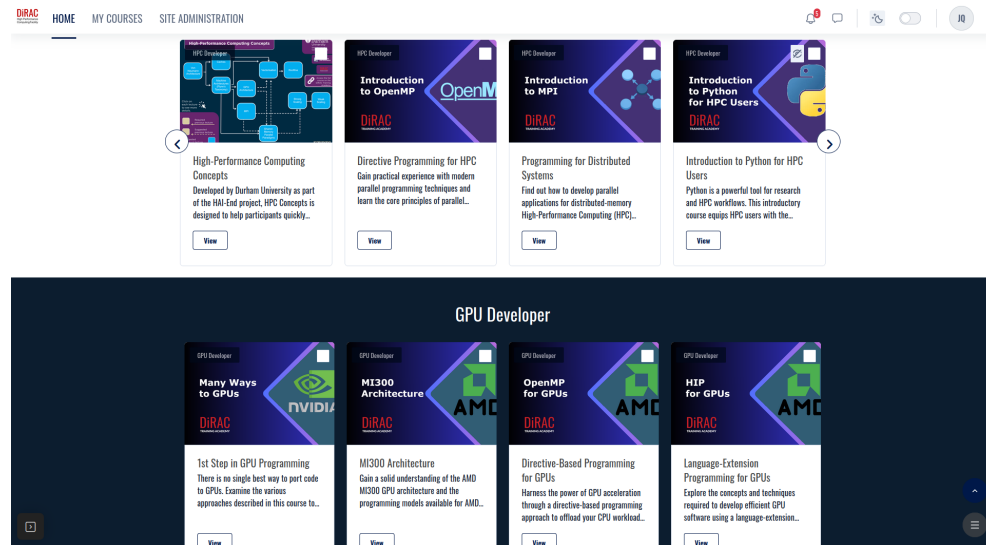
**What other training
is available?**

**Code clinic: Any
interesting
takeaways?**

**What other training
is available?**

DiRAC Training Academy

<https://training-academy.dirac.ac.uk/>



- Python for HPC
- MPI
- OpenMP
- Bash
- Slurm
- HIP
- Intro to Nvidia programming models
- OpenMP for AMD GPUs (transferable)

Upcoming DiRAC Training

- Intro to ML/AI
- GPPT self-paced courses:
 - **Holistic GPU training**
 - Profiling & performance
 - Software architecture & design
 - Testing, debugging & energy efficiency

& secret training hardware project

HPC Centres

Julich

Training materials of JSC courses

Here you can access **presentation slides**, **recordings**, **exercises** and **tutorials** from previous JSC training courses.

Tip: Get an initial overview of High-Performance Computing at JSC with the video recordings of the course "Introduction to Supercomputing at JSC – Theory & Practice".

Course title	Course material
Directive-Based GPU Programming with OpenACC	Slides
Erstellung wissenschaftlicher Texte mit LaTeX	Slides
ESM User Forum	Slides
High-Performance Computing with Python	Slides and exercises
High-Performance Scientific Computing in C++	Slides and exercises
In-Situ Visualization on High-Performance-Computers	Video recordings
Interactive High-Performance Computing with JupyterLab	Video recordings
Introduction to Machine Learning in the Application Area of Fluid Mechanics and Combustion Using HPC	Gitlab repository and shared notes
Introduction to Parallel Programming with MPI and OpenMP	Slides, exercises and tutorials

<https://www.fz-juelich.de/en/jsc/education/training-courses/training-materials>

EPCC

ARCHER2 Training

Details of all the face-to-face and online training courses and resources provided by EPCC as part of the ARCHER2 service.

Course Descriptions

An overview of the courses that are offered by EPCC as part of the ARCHER2 service.

Upcoming Training

Details of face-to-face and online training opportunities delivered by the ARCHER2 service.

Past Course materials

Repositories of materials from past ARCHER2 courses including videos from past online courses.

Course Locations

ARCHER2 training will be carried out at a range of venues across the UK.

Online Learning

Information on the ARCHER2 online learning, including the technology used and how to connect.

Webinars

Past Webinars and virtual tutorials

ARCHER2 YouTube channel

Includes videos of past Webinars and online training.

ARCHER2 Driving Test

The ARCHER2 Driving Test allows you to learn more about using ARCHER2 and get a small amount of ARCHER2 resource for your research.

Training Feedback

Provide feedback on ARCHER2 training courses.

<https://www.archer2.ac.uk/training>

Full Multi-GPU Course

The screenshot shows the GitHub interface for the repository 'tutorial-multi-gpu' by the user 'FZJ-JSC'. The repository is public and has 12 watchers, 82 forks, and 383 stars. The main content area displays a file tree with 11 folders and 1 file. The folders represent different parts of the course, including introductions, onboarding, and specific topics like MPI-Distributed Computing, GPU Parallelization, and NCCL/NVSHMEM. The file '.archive.mk' is also listed. The right sidebar contains an 'About' section with a description: 'Efficient Distributed GPU Programming for Exascale, an SC/ISC Tutorial'. It includes a list of related topics like cuda, exascale-computing, gpu, hpc, isc22, isc23, isc24, isc25, isc26, mpi, multi-gpu, nccl, nvshmem, sc21, sc22, sc23, sc24, sc25, and supercomputing. There are also links to the README, MIT license, and a 'Cite this repository' button. The 'Releases' section shows 10 releases, with the latest being 'ISC26 Tutorial' from 3 months ago.

Navigation: <> Code, Issues, Pull requests, Discussions, Actions, Projects, Security and quality, Insights

Repository: tutorial-multi-gpu (Public)

Stats: Watch 12, Fork 82, Star 383

Branches: main (3 Branches), Tags: 10 Tags

Search: Go to file

Buttons: Add file, Code

File/Folder	Description	Last Update
AndiH Fix time table ✓	293ce46 · 3 months ago	344 Commits
.etc	Update scripts for ISC26	3 months ago
.github/workflows	Add Github Actions	4 months ago
01-L_Introduction_Overview	Fix time table	3 months ago
01b-H_Onboarding	Update slides	3 months ago
02-L_Introduction_to_MPI-Distributed_Comp...	Updated Slides for ISC 2026	3 months ago
03-H_Multi_GPU_Parallelization	Update exercises header to ISC26	3 months ago
04-L_Performance_and_debugging_tools	Update slides for 04-L	3 months ago
05-L_Optimization_techniques_for_multi-GP...	Updated Slides for ISC 2026	3 months ago
06-H_Overlap_Communication_and_Comput...	Update exercises header to ISC26	3 months ago
07-L_Overview_of_NCCL_and_NVSHMEM_in_...	Updated Slides for ISC 2026	3 months ago
08-H_NCCL_NVSHMEM	Update exercises header to ISC26	3 months ago
09-L_CUDA_Graphs_and_Device-initiated_Co...	Update slides for 09-L	3 months ago
10-H_CUDA_Graphs_and_Device-initiated_Co...	Update exercises header to ISC26	3 months ago
11-L_Summary_Advanced	Update slides for SC25, Part 1	11 months ago
.archive.mk	--amend	3 months ago

About

Efficient Distributed GPU Programming for Exascale, an SC/ISC Tutorial

Related topics: cuda, exascale-computing, gpu, hpc, isc22, isc23, isc24, isc25, isc26, mpi, multi-gpu, nccl, nvshmem, sc21, sc22, sc23, sc24, sc25, supercomputing

Readme, MIT license, Cite this repository, Activity, Custom properties, 383 stars, 12 watching, 82 forks, Report repository

Releases 10

ISC26 Tutorial (Latest) 3 months ago

+ 9 releases

<https://github.com/FZJ-JSC/tutorial-multi-gpu>

AMD Developer Training

Training

Learn how to use AMD development platforms. Take self-paced training courses, find textbooks, sign up for webinars, and explore how-to resources.

AI Academy

Self-learning courses designed specifically for AI developers.

[Learn More >](#)

Adaptive Computing Learning Center

Learn how to design and program adaptive accelerators, SOMs, NICs, SoCs, and FPGAs using the AMD Vitis™ unified software platform and Vivado™ Design Suite.

[Visit Learning Center >](#)

Adaptive Computing YouTube

Access developer-to-developer content including training videos, tutorials, and other hands-on information.

[Watch Videos >](#)

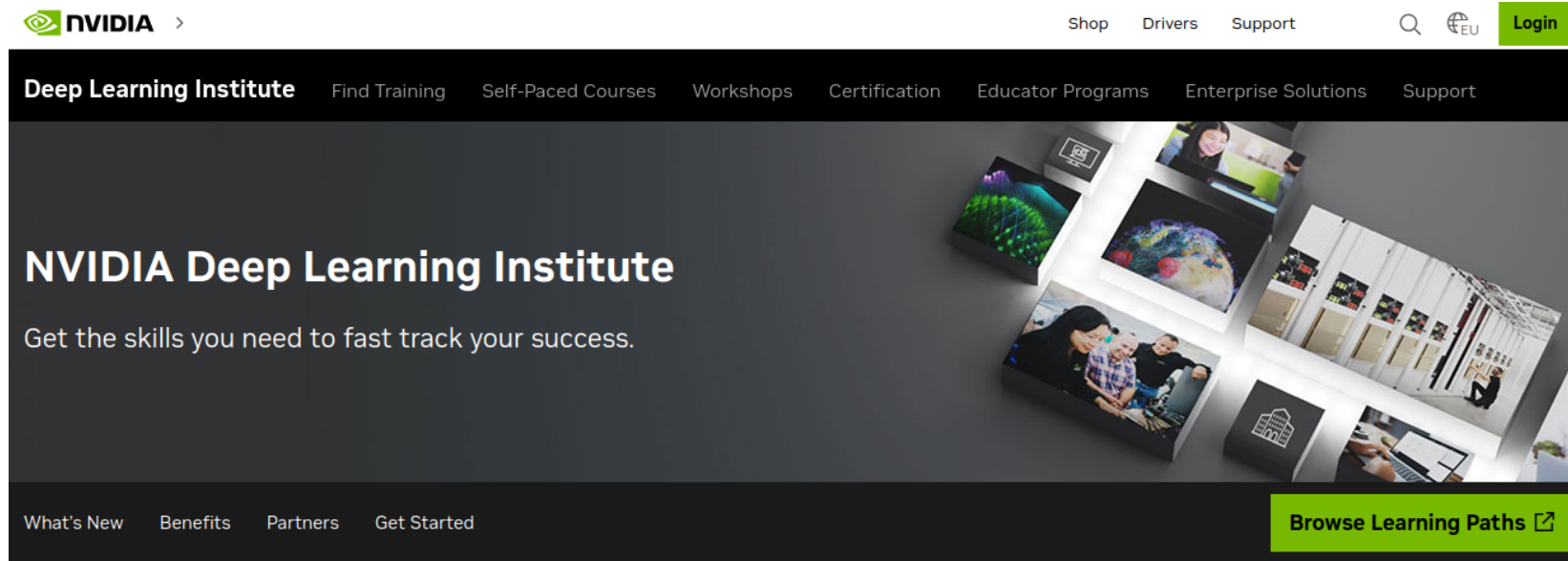
ROCm™ Training

Get the ROCm™ textbook, sign up for educational webinars, and more.

[Discover ROCm >](#)

<https://www.amd.com/en/developer/browse-by-resource-type/training.html>

Nvidia DLI



Take advantage of the latest technical training from NVIDIA and gain in-demand skills, hands-on experience, and expert knowledge in AI, data science, accelerated computing, and more.

<https://www.nvidia.com/en-eu/training/>
(Attend GTC online for free & access DLI courses)

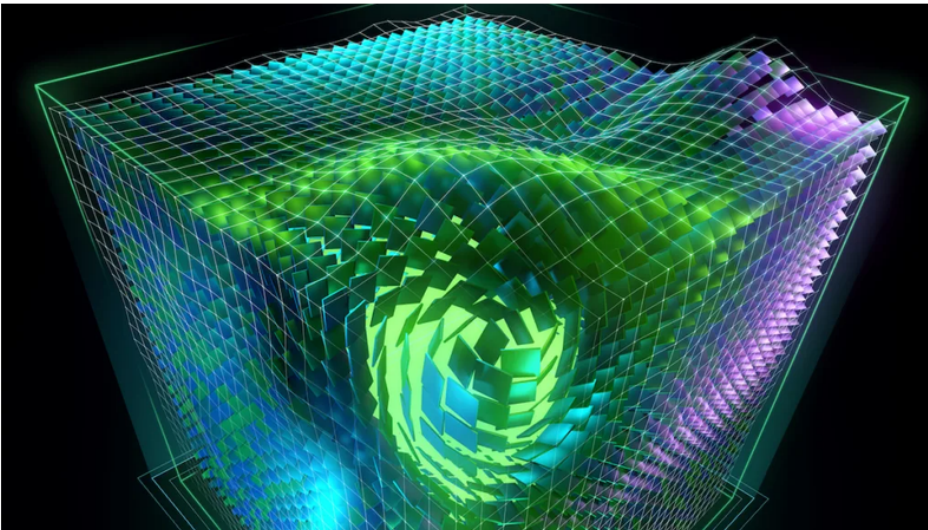
Blogs

Technical Blog [Subscribe >](#)

[Models](#) / [Libraries](#) / [Frameworks](#)

English ▾

Using Shared Memory in CUDA C/C++



Jan 28, 2013

 +55 Like  Discuss (36)

By [Mark Harris](#)

[https://
developer.nvidia.com/blog](https://developer.nvidia.com/blog)

 ROCm™ Blogs

Avoiding LDS Bank Conflicts on AMD GPUs Using CK-Tile Framework



[https://www.amd.com/en/
blogs.html](https://www.amd.com/en/blogs.html)

Other training

- Stanford's Parallel Computing
- 120 Days of CUDA Challenge
- Video series:
 - GPU concepts (great animations) (2024)
 - AMD HIP course (2024)
 - American UofBeirut's GPU course (2021)